MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, CHIP,
ESTABLISHED RELIABILITY
GENERAL SPECIFICATION FOR

This ammendment forms a part of MIL-R-55342E, dated 26 January 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

Delete table II (excluding footnotes) and substitute:

"TABLE II. Termination material.

Type	Material	Termination area	Code letters
Solderable 1/ Bondable Weldable	Gold	Wrap around 2/ One surface Bonding pads	G W 3/ P
Solderable 1/	Base metallization barrier metal, solder coated	Wrap around 2/	B
	Pretinned	Wrap around 2/ One surface	R 4/ S 3/
Bondable weldable	Platinum/gold	Wrap around One surface	U T 3/
Bondable	Palladium/silver or Platinum/silver	Wrap around	C
	Palladium/silver or Platinum/silver	One surface	D 3/

"

PAGE 23

3.20.3, delete and substitute:

"3.20.3 Supplying to higher failure rate levels. A manufacturer may supply to all higher failure rate levels than to which he is qualified. Parts qualified to lower failure levels, with procuring agency approval, are substitutable as long as the label reflects the original failure rate level ordered (see table VIII). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2)."

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:11Z
Check the source to verify that this is the current version before use.

TABLE VIII, delete and substitute:

"TABLE VIII. Failure rate substitution.

Failure rate	Failure rate substitute
S (.001)	.....
R (.01)	S
P (0.1)	S, R
M (1.0)	S, R, P

"

3.20.4, delete and substitute:

"3.20.4 Supplying to lower resistance temperature characteristics and or lower resistance tolerances. Parts qualified to lower resistance temperature characteristics and/or lower resistance tolerance levels are substitutable, with procuring agency approval, for higher resistance temperature or higher tolerance levels as long as the label reflects the original resistance temperature characteristic and/or resistance tolerance ordered (see tables IX and X). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2)."

PAGE 24

TABLE IX, delete and substitute:

"TABLE IX. Resistance tolerance substitution.

Resistance tolerance	Resistance tolerance substitute
0.1.....	.....
0.5.....	0.1
1.0.....	0.1, 0.5
2.0.....	0.1, 0.5, 1.0
5.0.....	0.1, 0.5, 1.0, 2.0
10.0.....	0.1, 0.5, 1.0, 2.0, 5.0

"

TABLE X, delete and substitute:

"TABLE X. Characteristic substitution.

Characteristic	Characteristic substitute
Y	.....
E	Y
H	Y, E
K	Y, E, H
M	Y, E, H, K

"

PAGE 27

4.4.4.2; delete the last sentence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:11Z
Check the source to verify that this is the current version before use.

Add the following new paragraphs:

"4.6.1.2.4 Subgroup 4 (solderability).

"4.6.1.2.4.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup 4 solderability test. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

"4.6.1.2.4.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use the following option to rework the lot:

Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.7.11. An additional 13 samples from each production lot shall be tested. If there are one or more defects in the production lot sample, that production lot shall be considered to have failed and that product will not be supplied to the requirements of this specification. Individual production lots that pass the solderability test are available for shipment.

"4.6.1.2.4.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract."

TABLE XII, add the following new subgroup:

"

Subgroup 4			
Solderability.....	3.17	4.7.11	see 4.6.1.2.4

"

TABLE XIII, delete and substitute:

"TABLE XIII. Sampling plans for PPM categories.

Lot size	Sample size
0 - 125	100%
126 - 3200	125
3201 - 10,000	200
10,001 - 35,000	315
35,001 - 150,000	500
150,001 - 500,000	800
500,001 - up	1250

"

TABLE XIV, delete subgroup 3 in its entirety.

4.6.1.3.1.3, delete in its entirety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:11Z
Check the source to verify that this is the current version before use.

PAGE 33

4.7.1.3(b), delete and substitute:

"b. Termination material "W" and "P" (weldable). Chip resistors with weldable terminations shall be mechanically attached to the test substrate and electrically connected by wire bonding flying lead interconnections between the chip termination and the test substrate metallized land areas. The interconnecting lead shall be 0.001 inch in diameter gold wire. The chip shall be mechanically mounted by any procedure which will not be the cause of or contribute to any failures of the chip resistor in any test."

PAGE 37

4.7.9, add the following new subparagraph:

"e. At the option of the manufacturer, the reverse sequence of table XVII may be as follows:

1 - Room temperature.
2 - Series of hot temperatures.
3 - Room temperature.
4 - Series of cold temperatures for the subgroup 1 tests."

PAGE 44

TABLE XX, style RM2512, under "will qualify style" column: Delete "RM0905" and substitute "RM2010".

Characteristic "Y", under "will qualify characteristic" column: Delete "E, Y, H" and substitute "Y, E, H".

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1215)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:11Z
Check the source to verify that this is the current version before use.